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Analyzing Open-Source eXtensible Business Reporting Language (XBRL) Financial Statement Data: A DuPont Analysis

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ABSTRACT: In this case, students will (1) develop an understanding of the data structure and format of financial statement data filed using the eXtensible Business Reporting Language (XBRL) framework; (2) extract, clean, and transform this XBRL data in order to conduct DuPont analysis; and (3) communicate the analysis findings in a professional written report that incorporates relevant visualizations. To complete this activity, students should make use of an appropriate process workflow automation tool, (i.e., Alteryx, Python, R), which facilitates the extract, transform, and load (ETL) process for large datasets in a structured manner. Although DuPont analysis is the focal point for this activity, students and instructors can answer a wide range of accounting and finance questions that leverage the XBRL data. As such, this activity serves as an introduction to XBRL and to process automation tools in general and may facilitate integration of analytics throughout the accounting or business school curricula.

Keywords: accounting analytics; XBRL; DuPont analysis; process workflow automation; Alteryx.

I. CASE ACTIVITY OVERVIEW

In this case activity, you will analyze a company (or multiple companies) using a popular analysis technique known as DuPont analysis. The financial statement data required to complete this activity is tagged and filed by companies using the eXtensible Business Reporting Language (XBRL) framework and is compiled by the Securities and Exchange Commission (SEC) for free public use. Your instructor will provide guidance regarding a specific technology application to use to prepare—extract, transform, and load (ETL)—the data for analysis. The analysis itself can be performed in a tool of your choice (i.e., PowerBI, Excel, Tableau), and the final deliverable will be a report discussing the results of your analysis with relevant data visualizations included. At the heart of this activity is the importance of establishing a repeatable (and perhaps even automated) process for performing the important ETL steps. For instance, the analysis is often performed on a predictable calendar (i.e., monthly, quarterly, or annually) as new data become

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available. Alternatively, a similar analysis may be performed across a variety of industries (i.e., when considering investing opportunities, analysts often want to consider a broad spectrum of companies and industries). Developing a repeatable and efficient ETL process is worthwhile in these and similar use cases so that the analyst has more of their time available for performing the analysis itself and effectively communicating the results. In the following sections, you will be introduced to the DuPont analysis technique, the XBRL data, and the concept of process workflow automation as it pertains to the ETL steps for this case activity. The case requirements and deliverable are found at the end of this document, and your instructor will provide guidance about the tools you should use and/or other resources that may be useful as you complete the activity.

II. ANALYZING COMPANIES USING THE BASIC DUPONT ANALYSIS TECHNIQUE

The basic DuPont analysis technique is a ratio analysis method that can be used to assess company performance, strategy, and risk across time and/or in comparison to other companies. Specifically, DuPont analysis relies on the following relationship model:

$$Return\ on\ Equity\ (ROE) = Profit\ Margin\ (PM) \times Asset\ Turnover\ (AT) \times Financial\ Leverage\ (LEV) \tag{1}$$

ROE presents a company's performance over a period of time relative to its equity (or book value) and is a useful tool for analysts to use to compare companies' profitability and expected return to shareholders. The three components of ROE decompose this performance measure into distinct elements: (1) company profitability (PM), (2) company operating efficiency (AT), and (3) company risk or financial leverage (FL). The specific calculations for each of these ROE measures is as follows, and Figure 1 provides an overview of the full model:

$$Return\ on\ Equity = Net\ Income\ /\ Average\ Total\ Equity\ decomposed\ \text{as}\ follows: \tag{2}$$

$$Profit\ Margin\ (PM) = Net\ Income \div Sales \tag{3}$$

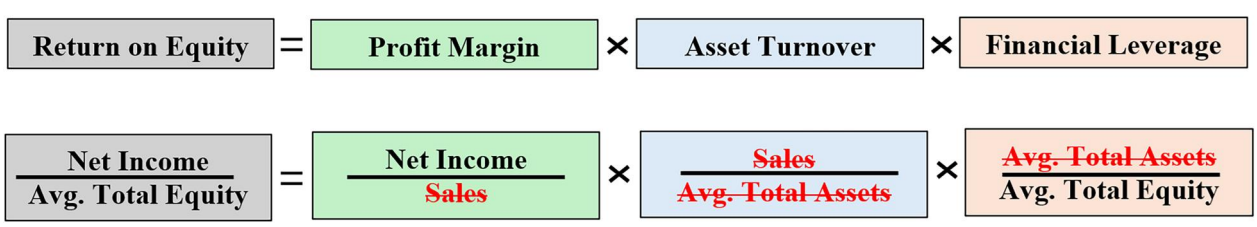
$$Asset\ Turnover\ (AT) = Sales \div Average\ Total\ Assets \tag{4}$$

$$Financial\ Leverage\ (FL) = Average\ Total\ Assets\ /\ Average\ Total\ Equity \tag{5}$$

DuPont analysis requires company financial statement data. Although it is useful for an analyst to calculate the DuPont ratio for a single company and year to better understand the components of ROE, this analysis is limited. Instead, in-depth analysis entails making comparisons across time and/or between companies, requiring data from multiple companies and/or years. Commonly, company-level financial statement analysis incorporates three types of data to facilitate a more comprehensive examination: (1) cross-sectional, (2) time-series, or (3) panel. ***Cross-sectional data*** span multiple companies for a single time period. These kind of data allow for comparative analysis of many different companies' performance or financial position during the *same* time period. These companies may be within the same industry or can be compared across industries. ***Time-series data*** follow a single entity or company over a period of many years. Time-series datasets are used to analyze changes in performance or financial position across a relevant period of time (i.e., quarterly, annually). ***Panel datasets*** combine these two data types, allowing for a robust analysis of many companies' performances or financial positions across time and within (or across) multiple industries. Given the

FIGURE 1

DuPont Analysis Formula Overview with Decomposition



(The full-color version is available online.)

data requirements for conducting these varying analyses, it is much more efficient for analysts to leverage technology to gather the required financial statement data than hand-collecting the data company-by-company and year-by-year. Researchers and professional analysts may be able to access data repositories such as Compustat or make use of tools such as Bloomberg terminals to access large company-level panel datasets for their analysis activities. However, these data resources are costly and may not be accessible to all individuals.

The SEC makes financial statement data available to the public for free and in a format that facilitates the use of large datasets for financial statement analysis. Specifically, the XBRL initiative requires that companies use a standardized data tagging taxonomy for the reported financial statement elements in their SEC filings. XBRL-tagged financial statement data are extracted from the financial statements and made available by the SEC in quarterly files for the company filings made during the relevant submission period. Using process automation tools and/or other ETL applications, analysts can work with these XBRL data to perform robust analysis at scale. In this activity, you will perform DuPont analysis for an industry of your choice for multiple time periods using the XBRL financial statement datasets sourced from the SEC's EDGAR database, posted on the SEC's website.

III. OVERVIEW OF THE DATA AND XBRL

XBRL is an important open-source standard for business reporting that facilitates large-scale data analysis. Initiated by the SEC in 2004 as a voluntary program, the first U.S. GAAP XBRL reporting taxonomy was compiled in 2008, and XBRL tagging was mandated beginning in 2009. In the years since, this taxonomy has been continuously evaluated and refined in an effort to facilitate comparability across companies and industries.[1] However, these data have important limitations to note. Given its recent adoption, XBRL-tagged data cannot be used for analysis before 2004, and the limited scale of adoption and inconsistency in tagging over time presents important challenges, most notably that many smaller companies did not consistently use XBRL tagging for many years after 2004. Further, many businesses continue to report elements not included in the standardized taxonomy (i.e. custom tags, referred to as extensions), suggesting that there remains a significant amount of noise in the data. In their review of XBRL, R. Hoitash, U. Hoitash, and Morris (2021) highlight that approximately one in six (~13–17 percent) tags are extended XBRL tags in filings spanning 2012 through 2019. Despite these limitations, XBRL presents a large and free dataset for public use in line with the SEC's goals to facilitate greater transparency in financial reporting. In addition, recent analysis has shown that XBRL data tagging has improved steadily beginning especially in 2019.[2]

Each quarter, the SEC publishes four financial statement data tables (**SUB**, **TAG**, **NUM**, and **PRE**) in a zip file at the following website: https://www.sec.gov/data-research/sec-markets-data/financial-statement-data-sets.

These data tables contain a variety of financial statement data associated with the registrant filings for the relevant quarter tagged using the XBRL taxonomy. Through the XBRL initiative and these publicly available data tables, analysts can obtain a wide range of financial statement data to conduct analysis for U.S. publicly traded companies across time. In addition to the data tables, each quarterly zip file includes an HTML document called **README** (i.e., a data dictionary), which helps facilitate data use. The document includes detailed information on the file formats and fields used in each data table.

The primary data tables within each quarterly zipped folder are contained in a tab-separated values (TSV) text file. This format makes it easy to extract the tagged data into programs like Alteryx, Excel, and Tableau for transformation and analysis. The first file to be familiar with is the **SUB** file. This file includes one record for each EDGAR submission filed with the SEC during the corresponding quarter (Q1 = January 1 through March 31; Q2 = April 1 through June 30; etc.). For reference, in each quarter of 2020, an average of about 15,000 filings were submitted, and these data are captured in the **SUB** data table. The number of 2020 submissions is considerably higher than in 2019, when approximately 7,000 to 9,000 filings were submitted each quarter. Companies have regularly filed annual (10-K) and interim/quarterly (10-Q) filings, which incorporate the XBRL taxonomy, but the usage in other filings (namely 8-K filings) has increased dramatically in more recent years. This activity will focus on data tagged within the annual 10-K filings (i.e., annual data). In summary, **SUB** provides overview information about the filer and the filing itself, including the filers' name, CIK code (SEC-assigned unique identifier), SIC code (industry classification), full address, filer status, fiscal year-end, and the type of filing and relevant period.

The second file, **NUM**, contains each of the tagged numerical elements in all of the filings submitted in a particular quarter. Each XBRL-tagged element is captured as its own row (i.e., a unique record) in the data table. In Q1 of 2020, **NUM** included just over 3 million rows of data. The subsequent three quarters included approximately 2.25 million

[1] XBRL International has several excellent resources available at the following website: https://www.xbrl.org/the-standard/what/
[2] See SEC XBRL custom tag trends at the following website: https://www.sec.gov/data-research/gaap-xbrl-custom-tags

rows of data each. These larger files present a challenge for Excel users, given the existing limits of approximately 1 million rows of data per sheet. It is also important to note that the Q1 **NUM** data tables are larger than they are in other quarters because the first quarter is when calendar year-end (December 31) companies submit their full annual reports (10-Ks).

Both **SUB** and **NUM** include an accession number (**adsh**) field. This field can be used as a unique identifier when matching company information from the **SUB** datafile to each of the tags in the filings and captured in the **NUM** table. The tag field is a unique identifier for each of the possible tags that companies can use. This allows users to filter for certain financial statement accounts. Additionally, because every number is tagged and given its own record, certain comparative financial statements result in duplicate tags. For example, a company that presents a two-year comparative balance sheet in its 10-K will have a record for total assets for both the current year and prior year, and these will be noted as two distinct records within the data table. Users of this information need to carefully filter the XBRL data to ensure their analyses are not confounded by the mistaken use of prior years' data.

The third file published by the SEC is **TAG**. This dataset provides detailed information about each tag. One of the unique identifiers used in **NUM** ("**tag**") connects to this table. This table can be used to search for tags that may be helpful to examine, but because the full tag name is provided in **NUM**, this dataset can mostly be ignored once relevant tags have been identified. The final file is **PRE**, which provides information about the presentation of each tag used by each filer. This includes the specific tag used, what financial statement the tag appeared on, the line order of the tag, and more. This data table provides important context for the analyst to consider comparability and data sourcing. Most analyses will largely rely on data from the **SUB** and **NUM** files, with these other data tables potentially used for assessing comparability, tagging consistency, and/or locating possible XBRL tags that might contain relevant numerical data. For this activity, the **SUB** and **NUM** files will be the primary data sources. A relationship chart of the XBRL data tables is provided in Figure 2.

DuPont analysis requires multiple periods of the SEC financial data tables to obtain relevant financial statement data for a chosen group of companies or industries. To help facilitate the analysis, process workflow automation tools can be used to extract the data from source data tables, join the extracted data, and then transform and clean the data to prepare it for analysis across multiple years. The next section provides a brief overview of process workflow automation and the ETL process for this case.

IV. OVERVIEW OF THE ETL PROCESS AND PROCESS WORKFLOW AUTOMATION

Raw data often must be extracted, transformed, and loaded into an appropriate tool for analysis. In this case, the raw XBRL data are comprehensive, but they are retrieved in a format that is not conducive to rigorous analysis. Further, the DuPont analysis variables must be created from the source financial statement data. Although it is possible

FIGURE 2

XBRL Relationship Table

Company filings are presented as a single record in the SUB table. For example, Microsoft's 10-K filing for a particular fiscal year would be displayed once.

Company filings will have several corresponding records in the NUM table. Each XBRL-tagged financial statement line item is its own record. For example, the financial statement line items like cash, inventory, accounts payable, and revenue will each be a distinct record.

In addition, in the NUM file, each distinct year presented in the financial statements is captured as a separate record. For instance, revenue for the current year and the prior two years is often presented (i.e., comparative financial information), and each of these three years of revenue data would be distinct records.

The SUB and NUM data tables are connected through a ***one-to-many inner join***, using the fields "adsh" and "ddate" from NUM. The resulting join output will maintain each NUM file record (i.e., cash, inventory, etc.), as noted in Note 2 above, but will now include relevant submission data from the SUB file.

A more detailed explanation and data dictionary can be found at https://www.sec.gov/aqfs.pdf

(The full-color version is available online.)

to perform the necessary ETL steps using a manual process, automated process workflows offer many advantages. The primary benefit of using an automated process workflow is that the automated workflows can be used to process large datasets that can be overwhelming in scope for manual processes. In this case, the XBRL **NUM** file includes all the tagged financial statement elements during a three-month submission period, comprising multiple million rows. Many manual processes cannot be easily applied to such large datasets. Another benefit of automated process workflows is that they can be repeated as data are updated, ensuring consistent application and analysis across time. Finally, automated process workflows provide clear documentation of the ETL steps performed on the data, facilitating accountability and transparency. Some of the process workflow automation tools require programming (i.e., Python, R, etc.), which may be a barrier without the requisite skills and/or extensive training. However, many professional accounting firms use process automation tools such as Alteryx (or similar), which offer a full suite of workflow automation capabilities, without requiring users to engage in coding.

To complete this DuPont analysis case, your instructor will guide you on the use of an appropriate process workflow automation application to prepare the XBRL financial statement data for analysis. Once this workflow is built, it can be used to efficiently facilitate additional analyses in the future integrating varying time periods, alternative company/industry filters, or a different selection of XBRL tags and financial statement data. Appendix A provides a conceptual guide to the ETL steps used to prepare the XBRL data for the required DuPont analysis.

V. DUPONT ANALYSIS ACTIVITY REQUIREMENTS

Your final deliverable for this activity is a professionally written report that presents the DuPont analysis for an industry (or industries) of your choice. In the report, you should provide a summary of the data and data sources used, a discussion of the ETL steps performed, and an overview of the chosen industry (or industries) to provide important background for your analysis. The focal point of your report should be the presentation of the DuPont analysis findings and conclusions reached. The report should thoughtfully craft a story/narrative and incorporate relevant visualizations to accentuate key points and frame the important takeaways. Specifically, this activity entails the following key steps:

1. Your instructor will provide you with resources that should help guide you through the process of extracting XBRL data and preparing the data for DuPont analysis. Using this guidance, create your own matching process workflow and attempt to replicate the final data output for the airline industry (SIC: 45) in a format similar to that illustrated in Appendix A.
2. Once you are confident that your process workflow appropriately extracts and prepares the example airline industry data, choose one or more different industries and modify the workflow appropriately to obtain the necessary DuPont analysis data for a new set of companies *and* for a time period of your choice. The final extracted data output should have at least five unique companies and two years of data (i.e., three years of data inputs).
3. Load the extracted data into the analysis tool of your choice. Perform a DuPont analysis of the data for your chosen industry, including relevant visualizations, and report on your findings as described above. Your analysis *may* include one or more of the following:
    a. Comparing a specific company to its industry peers (cross-sectional) or the industry averages (benchmarking) for ROE and each of the DuPont subcomponents.
    b. Examining year-over changes in ROE and each of the DuPont subcomponents for a selected company *or* for the industry as a whole.
    c. Comparing ROE and each of the DuPont subcomponents for a variety of companies across multiple industries.
    d. Examining how the financial leverage affects ROE for a selection of companies within or across industries.
    e. Highlighting the effect of differing business strategies (i.e., focusing on profit margin, asset turnover) on ROE for a selection of companies within or across industries.

REFERENCES

Hoitash, R., U. Hoitash, and L. Morris. 2021. eXtensible business reporting language (XBRL): A review and implications for future research. *Auditing: A Journal of Practice & Theory* 40 (2): 107–132. https://doi.org/10.2308/AJPT-2019-517

INDEX OF SUPPLEMENTAL MATERIALS

SIC Codes - Final.xlsx

APPENDIX A

Process Workflow Conceptual Steps

The following are conceptual, high-level steps that will be performed on the XBRL data to prepare the data for the analysis part of the case. Recommended ETL applications include Alteryx, R, SQL, Tableau Prep, Microsoft Power Query, or Python. Tools that focus on data visualization only such as Tableau Desktop and Qlick are not recommended as they are not optimized for extraction and transformation of data (i.e., ETL). Depending on the tool used, the steps and/or order of operations may vary. But, overall, the goal is to produce a final export data file formatted similar to that shown in the "After Example" below.

Conceptual ETL Steps

- Download the data from the SEC website using the link below. The data will download as .txt files. You will need at least two years of data for the analysis, and two years of input data will generate one year of DuPont analysis data (because of the averages incorporated in the computations). If two years of DuPont analysis data are desired, three years of input data will be needed.
  - Link: https://www.sec.gov/dera/data/financial-statement-data-sets.html
- Import the SUB and NUM data .txt files into the tool of choice. Once imported, do a review of the data and ensure that categorical and numerical data imported correctly. Slight modifications might need to be made. For example, make sure that years are categorical (i.e., 2024) and do not appear as numerical data (i.e., 2,024).
  - SUB Data:
    - Apply the following filters to keep all 10-K data for accelerated filers.
      - "form" = "10-k"
      - "AFS" = "1-LAF" or "2-ACC"
    - Convert the "period" into a date field in "YYYYMMDD" format.
  - NUM Data:
    - Apply the following filters to filter out coregistrants, filter out segment data, and display units of measures in U.S. dollars. Finally, filter the tags to only include relevant items for DuPont.
      - "coreg" = " " (is null, blank, etc.)
      - "segments" = " " (is null, blank, etc.)
      - "uom" = "USD"
      - "tag" = "Assets," "NetIncomeLoss," "Revenues," "StockholdersEquity," "RevenueFromContractWithCustomerExcludingAssessedTax," "NetIncomeLossAvailableToCommonStockholdersBasic"
    - Convert the "ddate" into a date field in "YYYYMMDD" format
- Import the two-digit SIC code file (SIC Codes - Final.xlsx) into the tool of choice.
- Join the SUB and NUM data together using an inner join using the "adsh" fields and the new date fields created in both the SUB and NUM tables.
- Create a new "sic2" code that is the first two digits of the "sic" field within the dataset.
- Join the resulting table to the two-digit SIC code file using an inner join using the "sic2" field and the "code value" field from the two-digit SIC code file.
- Using the dataset with all fields, cross-tabulate (or transpose) the data on all of the fields except for the Tags and Values. The end result should have all the tags for one year on a single line.
  - Before Example: There will be four records for each company and year combination and one for each "tag."

(continued on next page)

APPENDIX A (continued)

Company	Period	Tag	Value	Attribute A	Attribute B	Attribute C
BOEING CO	20X21231	Assets	138,552,000,000	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	NetIncomeLoss	(4,202,000,000)	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	Revenues	62,286,000,000	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	StockholdersEquity	(14,999,000,000)	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	Assets	58,158,000,000	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	NetIncomeLoss	(11,873,000,000)	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	Revenues	58,158,000,000	37	Transportation Equipment	Manufacturing
BOEING CO	20X21231	StockholdersEquity	(18,316,000,000)	37	Transportation Equipment	Manufacturing

- After Example: There will be one record for each company and year combination, and all the tags will be attributes in the record.

Company	Attribute A	Attribute B	Attribute C	Period	Assets	NetIncomeLoss	Revenues	Stockholders Equity
BOEING CO	37	Transportation Equipment	Manufacturing	20X21231	138,552,000,000	(4,202,000,000)	62,286,000,000	(14,999,000,000)
BOEING CO	37	Transportation Equipment	Manufacturing	20X21231	58,158,000,000	(11,873,000,000)	58,158,000,000	(18,316,000,000)

- Sort the data on "cik" and "Fy_focus" descending.
- Create new fields needed for the DuPont analysis using "Tag" fields in the dataset:
  - **Final Revenue** = "Revenues." If "Revenues" is blank, use the value in "RevenueFromContractWith CustomerExcludingAssessedTax."
  - **Final Income** = "NetIncomeLoss." If "NetIncomeLoss" is blank, use the value in "NetIncomeLoss AvailableToCommonStockholdersBasic."
  - **Average Assets** = "Assets." To get the average, you must use a multirow calculation to add the assets in the current year to the assets of the prior year and divide by 2. The calculation should only be performed if the "cik" field in the current row and the row beneath is the same, designating that it is the same company.
  - **Average Equity** = "StockholdersEquity." To get the average, you must use a multirow calculation to add the "StockholdersEquity" in the current year to the StockholdersEquity of the prior year and divide by 2. The calculation should only be performed if the "cik" field in the current row and the row beneath is the same, designating that it is the same company.
- Apply a filter to keep only records that have values in all four of the previously created fields.
- Create new fields to support the DuPont:
  - **Net Profit Margin** = Final Income / Final Revenue
  - **Asset Turnover** = Final Revenue / Average Assets
  - **Financial Leverage** = Average Assets / Average Equity
  - **Dupont** = Net Profit Margin × Asset Turnover × Financial Leverage
- Filter field "sic2" to "45," which represents the airline industry.
- Extract the results to Excel or into another file format of your choosing for analysis. Note that the financial statement line items are likely to include large values, and, as such, they might display in scientific notation in certain tools (i.e., Excel). It is important to review the data and ensure that the values are appropriately formatted for meaningful presentation of the results.

VI. CASE LEARNING OBJECTIVES AND IMPLEMENTATION GUIDANCE

Case Purpose and Related Literature

In 2013, the Association to Advance Collegiate Schools of Business (AACSB) approved a new technology standard (A7) for accounting program accreditation. In an updated set of accounting accreditation standards passed in 2018, the technology standard was reaffirmed (now as A5) with two important modifications from the original standard: (1) the 2018 standard emphasizes student *and* faculty technology competencies, and (2) the 2018 standard de-emphasizes data analytics concepts and definitions in favor of facilitating hands-on technology skills development (AACSB 2018). This teaching case provides students and instructors an opportunity to develop their technology competency using a hands-on activity in the spirit of the accreditation standards and reflecting the evolving professional accounting landscape. In this activity, XBRL financial statement data will be used to perform DuPont analysis. The analysis itself is relatively straightforward, but leveraging the large, publicly available XBRL dataset will require students and instructors to develop technology skills and data competencies related to financial accounting data.

XBRL data are provided on a quarterly basis by the U.S. Securities and Exchange Commission (SEC). These quarterly data tables contain all of the XBRL-tagged data elements from a variety of SEC filings submitted during the corresponding quarter. A number of recent academic articles discuss the benefits and limitations of using XBRL in the classroom and for research (R. Hoitash and U. Hoitash 2018; R. Hoitash, U. Hoitash, and Morris 2021; Du, Huddart, and Jiang 2023). According to Du et al. (2023), the as filed XBRL data have three key advantages over financial statement data provided by aggregators such as Compustat, Bloomberg, or FactSet: (1) as filed XBRL data are more granular than the often-standardized aggregators' data, (2) the as filed XBRL data are verifiable and reproducible because it adheres to an authoritative public taxonomy, and (3) the as filed data are available immediately once filings are submitted to the SEC.

The SEC also distributes aggregated as filed XBRL data via quarterly data tables that can be accessed free of charge by the public. The data tables are large and likely require cleaning and transformation steps to be taken before the data can be reliably analyzed. As such, process workflow automation tools should be used to facilitate transparent and repeatable extract, transform, and load (ETL) activities to prep the XBRL data for analysis.

We encourage using Alteryx Designer to complete the necessary ETL activities. Alteryx is an intuitive and powerful tool for working with large datasets such as the XBRL data. Further, Alteryx is rapidly being adopted by accounting professionals, so experience with this software benefits students as they prepare for their careers in public accounting. The scope of the financial reporting data used in this case is relatively large (e.g., the quarterly **NUM** files contain more than 1 million rows of data each), and the analysis requires accessing relevant data situated across multiple data tables, such that data preparation and cleaning (i.e., ETL steps) must be performed before analyzing the data. Analysis applications, such as Alteryx, allow for the creation of a clearly documented and repeatable workflow to prepare the data for analysis. However, instructors can integrate any process workflow automation tool they prefer students use (i.e., R, Python, PowerBI, Tableau Prep Builder, etc.) for the ETL steps, dependent on specific course learning objectives and/or available technology resources. We provide a conceptual guide to the required ETL steps as part of the main teaching case to aid instructors and students in developing the process workflow in whichever application they choose. However, we also provide a step-by-step (Alteryx) instruction guide (XBRL Dupont Instruction Guide - CA.docx) along with the supplementary resources and encourage the distribution and use of this resource for those who choose to leverage Alteryx for the required ETL work. The analysis activity mapped out in this case is straightforward, but it showcases the breadth of the available financial statement data made available by the SEC as well as some of the challenges inherent when using these tagged data.

Several recent academic cases or teaching resources facilitate the integration of XBRL data and process automation tools—such as Alteryx—in accounting courses and curricula. Tadesse and Vincent (2022) introduce students to XBRL while also developing data analytics and ETL skills. In the case, students extract and combine large datasets to perform analysis and provide feedback to a supervisor regarding the selection of an XBRL software solution. This case aids students in building a foundational knowledge of the XBRL taxonomy and presents an opportunity to reflect on the internal decision-making processes companies may undertake as they move to implement XBRL tagging in their filings. Our case differs from this case in that it incorporates multiple XBRL tags and data elements over a multiyear time period.

In response to the widespread adoption of Alteryx in accounting practice, a number of recent academic resources focus on this tool in teaching cases (Cheng and Varadharajan 2021; Cheng, Rhoades-Catanach, and Watson 2023) or as an implementation guide for instructors considering integrating Alteryx in an Accounting Information Systems course (Zhang and Stone 2023). Cheng and Varadharajan (2021) implement an Alteryx process workflow to calculate net migration to/from different states as part of a tax policy analysis activity, and Cheng et al. (2023) leverage Alteryx

to create a modified accelerated cost recovery system (MACRS) depreciation workflow in a tax setting. Both teaching cases serve as effective Alteryx resources for use mainly in tax or analytics courses. Our case leverages Alteryx to obtain and analyze XBRL financial statement data, which could be integrated in a variety of undergraduate and graduate courses, including all levels of financial accounting (Principles, Intermediate, and/or Advanced), as well as Auditing, Data Analysis, and/or Financial Statement Analysis. As the use of XBRL data in teaching and research applications increases, we believe that Alteryx (and similar process automation tools) will be extremely useful for accessing and making use of the data. As such, this case should be a great introductory resource for students and faculty alike who are interested in harnessing the full potential of the XBRL data.

Learning Objectives

This full case activity requires students to navigate the entire data analysis cycle. Students will perform a DuPont analysis for an industry of their choice using a large, publicly accessible dataset composed of financial statement data extracted from company filings using the XBRL taxonomy. To successfully complete this activity, students will need to be able to pose their own questions, develop a working competency of the relevant data, perform integral data analysis steps, including the ETL activities, and communicate their work and findings effectively in a written report that incorporates visualizations. Although this case is intended to incorporate the full data analysis cycle, instructors may choose to modify the case to focus on a subset of the following objectives, as appropriate for the course setting, student demographics, and identified course learning objectives.

Data Competency and XBRL-Related Objectives

After completing this activity, students should be able to:

1. Describe what XBRL is and why it might be used by investors and analysts (Q1)
2. Explain XBRL tags and the general reporting framework for XBRL (Q2)
3. Use XBRL-extracted data to create a DuPont analysis (Q1–Q5)

Data Analysis-Related Objectives

After completing this activity, students should be able to:

1. Extract, transform, and prepare XBRL data for DuPont analysis (Q3)
2. Modify an existing process automation workflow to construct the measures for a DuPont analysis on a chosen industry (Q4)
3. Leverage DuPont analysis to examine company performance and financial position within a chosen industry across time (Q5)

Communication Objective

After completing this activity, students will have improved their ability to:

1. Create effective visualizations that communicate the analysis results to an appropriate audience (Q6)

VII. IMPLEMENTATION GUIDANCE

Target Audience and Required Preparation

This teaching case is intended to be used in an accounting analytics course and has been used in both undergraduate and graduate settings. The case will likely be most effective after students have obtained a basic understanding of XBRL and have been introduced to process workflow automation. This case could be used as an introduction to both XBRL and process workflow automation and could be integrated in any course that involves financial accounting and/or financial reporting (i.e., Intermediate Accounting, Financial Statement Analysis, etc.). This course could also be used in Auditing or Accounting Information Systems courses, perhaps with modified analysis activities, and the case would work for both graduate and undergraduate students. Many of the students who completed this case did not have prior experience with Alteryx (the process automation application the authors used to facilitate this case), and several students did not have much prior knowledge about XBRL before working on this case.

Alteryx Resources

We encourage using Alteryx to facilitate completing the required ETL steps for this case. Alteryx is free to use for educators and students. However, at the time of writing, it is only available for installation and use on Windows/PC devices. It is possible to run Alteryx using a Mac and Windows Parallels or Boot Camp, but, in our experience, many students do not have these accessibility tools installed on their devices. Alteryx also released a cloud version (Designer Cloud) that would be suitable to complete the case. The overall functionality is limited compared with the desktop version, but the necessary tools for this case are available with minimal modifications to the step-by-step guide (XBRL Dupont Instruction Guide - CA.docx). In the courses where this case was adopted (including for the efficacy testing), Alteryx was deployed on both physical and virtual lab computers. Students would complete their work during class time (i.e., flipped class) or could access the lab machines (whether physical or virtual) outside of class. In our experiences, the virtual lab computers were a great option because students could access the necessary tools anywhere they had a reliable internet connection. However, working within a virtual lab setting may entail slower processing time given the required internet connection or the configuration and capacity of the virtual lab environment.

We encourage instructors to create their own lessons that cover the CORE Alteryx topics or have students complete free Alteryx lessons directly from the Alteryx community. For instance, we have used an XBRL tag finding activity to introduce students to Alteryx and to the available XBRL tags for a variety of companies. This activity (Tag Finder Worksheet - Activity.docx; included in the supplementary materials and described further below) is also a useful way for students to think about the larger DuPont analysis activity systematically by identifying needed data and finding that data in the various company filings. In addition, students and instructors may find the available Alteryx community resources useful. The resources are only available to members, so instructors and students are required to create an account and join the community before accessing the lessons, tutorials, and other resources.[3] The following CORE lessons are recommended (approximate length):

CORE items (roughly 65 minutes in total)

- Introduction to Alteryx Designer (4 minutes)
- Understanding Data Types (9 minutes)
- Inputting Data into Your Workflow (6 minutes)
- Viewing Data (5 minutes)
- Filtering Data (7 minutes)
- Joining Data (6 minutes)
- Sorting Data (5 minutes)
- Summarizing Data (5 minutes)
- Diving into Expressions (11 minutes)
- Writing Data (5 minutes)

VIII. EVIDENCE OF CLASSROOM EFFICACY

This case was administered in one graduate accounting analytics course with 50 students at a regional public university in the northeast and in one undergraduate accounting analytics course with 32 students at a regional private university in the northwest. Pre- and post-case surveys were approved by the institutional review board and are formatted similar to previous cases (Dow, Watson, and Shea 2013; Cunningham and Stein 2018). The pre-case survey was administered before the case materials were shared with the students. Students were informed that the survey would evaluate their knowledge and comfort levels with certain course-related topics and case learning objectives. As tested in class, students were required to work with Alteryx as the process workflow automation tool for data prep. Once the data were prepped, students could use analysis tools of their choice to complete the DuPont analysis. After completion of the full case activity, students were invited to complete a post-case survey. This survey included the same questions as the pre-case survey, as well as additional perception-based questions to measure case effectiveness. The students were informed that completion of the surveys was both voluntary and anonymous and would have no impact on their course grade.

Of the 82 students who completed the pre-survey, 78 students (95.1 percent) completed the post-case survey. The drop-off in participants was attributed to class absence on the date the post-case survey was administered. Table 1 presents the results of the six questions related to the key learning objectives broken out by undergraduate and

[3] The Alteryx resources can be accessed here under the "All Lessons" header (account required): https://community.alteryx.com/t5/Interactive-Lessons/tkb-p/interactive-lessons

TABLE 1

Student Efficacy: Pre- and Post-Case Questions

Pre- and Post-Case Questions	Pre-Case Mean (Std. Dev.)			Post-Case Mean (Std. Dev.)			Mean Difference		
	UG n = 32	Grad n = 50	Total n = 82	UG n = 31	Grad n = 47	Total n = 78	UG	Grad	Total
Q1: I can describe what XBRL is and how XBRL data can be used in data analysis	1.44 (0.62)	2.38 (1.09)	2.01 (1.03)	3.61 (0.62)	3.87 (0.77)	3.77 (0.72)	2.18***	1.49***	1.76***
Q2: I understand XBRL tagging	1.34 (0.55)	2.24 (1.15)	1.89 (1.05)	3.52 (0.72)	3.87 (0.85)	3.73 (0.82)	2.18***	1.63***	1.84***
Q3: I am confident in my ability to import and clean data in Alteryx	1.59 (0.84)	2.46 (1.07)	2.12 (1.07)	3.84 (0.78)	3.87 (0.88)	3.86 (0.83)	2.24***	1.41***	1.74***
Q4: I am confident in my ability to modify an existing Alteryx workflow to fit my needs	1.53 (0.72)	2.36 (0.98)	2.04 (0.97)	3.65 (0.80)	3.89 (0.79)	3.80 (0.80)	2.11***	1.53***	1.76***
Q5: I am confident in my ability to use Alteryx to analyze a large dataset	1.53 (0.72)	2.28 (0.95)	1.99 (0.94)	3.65 (0.76)	3.62 (0.85)	3.63 (0.81)	2.11***	1.34***	1.64***
Q6: I am confident in my ability to visualize and communicate analytic results	2.56 (1.41)	2.82 (1.02)	2.72 (1.19)	4.16 (0.69)	4.09 (0.83)	4.12 (0.76)	1.60***	1.27***	1.40***

*** Indicates $p < 0.001$.

graduate. Table 2 presents the results of the additional perceptions-related questions for undergraduate and graduate students, respectively.

The combined results in Table 1 indicate that the case was effective in achieving the desired outcomes. Responses in Panel A were measured on a five-point Likert scale, where 1 represents "strongly disagree" and 5 represents "strongly agree." Both graduate (Grad) and undergraduate (UG) students saw a significant increase between pre- and post-case questions.

Responses in Table 2 were provided on a five-point Likert scale, where 1 represents "strongly disagree" and 5 represents "strongly agree." The survey results were relatively consistent for both classes measured. Students felt that the case was beneficial to their learning (Q1 total mean of 4.45), was the appropriate rigor (Q2 total mean of 4.38), and required critical thinking (Q3 total mean of 4.5). Especially relevant to accounting analytics, students noted that the activity improved their confidence in working with large datasets (Q4 total mean of 4.39). The mean values for each of the questions were significantly different (p-value = 0.001) from the scale's midpoint value of 3 using one-sample t-tests.

Qualitative responses from the students also suggested that the case had achieved the learning objectives. Students found the case challenging but enjoyable, and several students noted that it greatly improved their confidence in working with data analysis tools and large datasets. Students also noted that they enjoyed the flexibility of the case in that it allowed them to analyze an industry of their choice over a flexible time period. Some quotes from students were as follows:

> I think that the overall entire activity was beneficial to my learning. The case provided me with a good understanding of how to take the raw data and organize and adjust it so it can be effectively analyzed.

> The case was interesting to do and certainly challenging. I felt like I was at work where I'm expected to learn and use new tools.

> Having the opportunity to dive into XBRL and Alteryx was enjoyable. I can see XBRL being important in my career. I thoroughly enjoyed using Alteryx.

> Although I was hesitant, I really enjoyed this activity and I am glad that we had to use Alteryx and XBRL data.

TABLE 2

Student Experience: Post-Case Only Questions

	Grad Mean (Std. Dev.) Total n = 31	Grad Mean (Std. Dev.) Total n = 47	Total Mean (Std. Dev.) Total n = 78
Q1: The activity was beneficial to my learning	4.13*** (0.81)	4.66*** (0.52)	4.45*** (0.70)
Q2: The activity was the appropriate level of difficulty for this course	4.16*** (0.69)	4.53*** (0.58)	4.38*** (0.65)
Q3: The activity required me to use critical thinking skills	4.19*** (0.70)	4.70*** (0.46)	4.50*** (0.62)
Q4: The activity improved my confidence in being able to work with large datasets	4.23*** (0.67)	4.51*** (0.66)	4.40*** (0.67)

*** Indicates $p < 0.001$.

IX. CASE MODIFICATIONS AND OPTIONAL EXTENSIONS

Optional Reading Assignment

Depending on the course and student demographics, instructors may choose to leverage this case to integrate XBRL more holistically within the course. In these settings, instructors may require students to read and reflect on some additional materials, which provide an overview of XBRL or the utility of the XBRL data. We have found the following readings particularly useful and encourage they be assigned for students to read as appropriate:

- **XBRL overview (suitable for undergraduate and graduate students):** Hoitash et al. (2021).
- **Use of XBRL in an academic research context (suitable for graduate students):** Hoitash and Hoitash (2018).

In addition, Tadesse and Vincent (2022) published a case that exposes students to the ETL process and XBRL data (The ViewDrive Case). This case exposes students to the XBRL taxonomy and associated decision-making. The ViewDrive Case also incorporates ETL (joining two large datasets, including XBRL data) and analysis, so this case may serve as an effective foundation or precursor activity to this case. Note that students were not required to complete any required readings about XBRL nor were they assigned the ViewDrive Case for the case implementation associated with the efficacy results shared in the prior section. However, the resources referenced above have been extremely useful in designing and implementing this case specifically and in motivating the use of XBRL as an academic resource to be integrated in accounting courses more generally.

Optional Tag Finder Activity

Instructors have the option of supplementing the main case with a pre-activity that has students investigate the XBRL tags used by a company of their choice. This will allow students to have a better appreciation for the various tags that companies are using to represent their financial statement line items. After downloading the data, and building the workflow (Tag Finder Worksheet - Activity.docx), students will be presented with all of the tags used by their company of choice. Students are then prompted to identify relevant tags for their analysis requirements by comparing the tags to the company's 10-K reports. Instructors are provided with a worksheet that students can use to document their company-specific findings. This worksheet is provided in the supplemental materials for this case. Once the exercise is completed, students can compare and contrast their findings and see how XBRL tags might differ from company to company. Through this activity, students also will be introduced to process automation as they leverage a very simple, repeatable workflow to complete this task. We provide supplemental resources that present this activity using Alteryx, but alternative tools can be used (including Tableau Desktop), depending on instructor preferences. The activity serves as a great introduction to both XBRL and Alteryx, and it may be useful for students to complete before they are given the full case.

X. OPTIONAL VISUALIZATION CRITIQUE AND EDIT ACTIVITY

In certain course settings, this case may be assigned after students have completed a unit on visualizations, and, if so, students should be able to think through effective visualization techniques for presenting the results of the DuPont analysis for the chosen industry and time period. However, we did not provide example visualizations specific to DuPont analysis before this case so as to encourage greater variety and creativity in student responses. As a result, the submitted visualizations in the final deliverable varied considerably from student-to-student, with some being very effective and others being quite limited or marginal in their effectiveness. One possible extension of this case is to provide specific review feedback or critique on the visualizations submitted and to offer students a chance to revise their visualizations as a follow-up activity. The feedback could be provided by the instructor or as a peer-review activity and would encourage a more immersive application of data visualization at the instructor's discretion.

TEACHING NOTES

Teaching Notes are available only to full-member subscribers to the *Issues in Accounting Education* through the American Accounting Association's electronic publications system at https://publications.aaahq.org. Full-member subscribers can use their usernames and passwords for entry into the system where the Teaching Notes can be reviewed and printed. Please do not make the Teaching Notes available to students or post them on websites.

Teaching Notes for this case are available here.

If you are a full member of AAA with a subscription to *Issues in Accounting Education* and have any trouble accessing this material, please contact the AAA headquarters office at info@aaahq.org or (941) 921-7747.

REFERENCES

AACSB. 2018. 2018 Eligibility Procedures and Accreditation Standards for Accounting Accreditation. Tampa, FL: AACSB. https://www.aacsb.edu/-/media/documents/accreditation/accounting/standards-and-tables/2018-accounting-standards.pdf?rev=a2f3f6a396c24b0eb357b19a41c6cba8&hash=965FED94C79C756976B038B5F2151591

Cheng, C., and A. Varadharajan. 2021. Using data analytics to evaluate policy implications of migration patterns: Application for analytics, AIS, and tax classes. *Issues in Accounting Education* 36 (2): 111–128. https://doi.org/10.2308/ISSUES-19-098

Cheng, C., S. Rhoades-Catanach, and L. Watson. 2023. Data analytics for undergraduate tax students: An Alteryx case study for MACRS depreciation. *Issues in Accounting Education* 38 (4): 145–164. https://doi.org/10.2308/ISSUES-2022-005

Cunningham, L. M., and S. E. Stein. 2018. Using visualization software in the audit of revenue transactions to identify anomalies. *Issues in Accounting Education* 33 (4): 33–46. https://doi.org/10.2308/iace-52146

Dow, K. E., M. W. Watson, and V. J. Shea. 2013. Understanding the links between audit risks and audit steps: The case of procurement cards. *Issues in Accounting Education* 28 (4): 913–921. https://doi.org/10.2308/iace-50508

Du, K., S. Huddart, and X. D. Jiang. 2023. Lost in standardization: Effects of financial statement database discrepancies on inference. *Journal of Accounting and Economics* 76 (1): 101573. https://doi.org/10.1016/j.jacceco.2022.101573

Hoitash, R., and U. Hoitash. 2018. Measuring accounting reporting complexity with XBRL. *The Accounting Review* 93 (1): 259–287. https://doi.org/10.2308/accr-51762

Hoitash, R., U. Hoitash, and L. Morris. 2021. eXtensible business reporting language (XBRL): A review and implications for future research. *Auditing: A Journal of Practice & Theory* 40 (2): 107–132. https://doi.org/10.2308/AJPT-2019-517

Tadesse, A. F., and N. E. Vincent. 2022. Combining data analytics with XBRL: The ViewDrive Case. *Issues in Accounting Education* 37 (1): 197–215. https://doi.org/10.2308/ISSUES-2020-048

Zhang, C., and D. N. Stone. 2023. Integrating Alteryx designer and Tableau desktop into the AIS course: An analytics mindset model. *Issues in Accounting Education* 38 (2): 35–61. https://doi.org/10.2308/ISSUES-2021-103

INDEX OF SUPPLEMENTAL MATERIALS

XBRL Dupont Instruction Guide - CA.docx
Tag Finder Worksheet - Activity.docx

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